

Total marks: 30

Duration: 1.5 hours

Answer 3 out of 4 questions

- 1(a): Derive Cauchy-Riemann equations which provide the necessary condition in a function $f(z)$ to be analytic.
 (b) Derive the polar form of Cauchy-Riemann equations. [6 + 4]

- 2(a) Prove Cauchy's theorem $\oint f(z)dz = 0$, where $f(z)$ is an analytical function in a region C .
 (b) Prove Cauchy's integral formula for n^{th} derivative,

$$f^n(a) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z-a)^{n+1}} dz$$

[5+5]

- 3(a) Find $\oint_C \frac{z^2 e^z}{z+i} dz$ where the curve C is (i) $|z| = \frac{1}{2}$, (ii) $|z| = 2$, (iii) $|z| = 1$
 b) Find $\oint \frac{e^z}{(z-1)(z+3)^2} dz$, where curve C is $|z| = \frac{3}{2}$

[6+4]

- 4 (a) Discuss various types of singularities of a complex function $f(z)$ using Laurent's series.
 (b) $\oint \frac{1-3z}{z^2(1-z)^2(z-2)}$, where $C : |z| = \frac{3}{2}$
 (c) $\oint \frac{z^2-2z}{(z+1)^2(z^2+4)}$ where $C : |z| = 2$

[2+4+4]